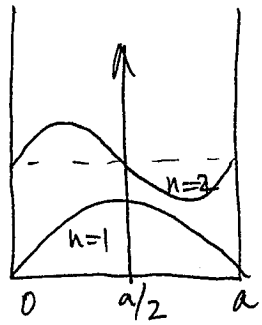
P6.1 (a)



$$H' = \alpha\,\delta(x - a/2)$$

Unperturbed problem:

$$\psi_n = \sqrt{\frac{2}{a}}\,\sin\left(\frac{\pi n}{a}x\right) \qquad E_n = \frac{\pi^2\hbar^2 n^2}{2ma^2}$$

$$E_n^1 = \langle \psi_n | H' | \psi_n \rangle = \int_0^a \sqrt{\frac{2}{a}}\,\sin\left(\frac{\pi n}{a}x\right)\,\alpha\,\delta(x-a/2)\sqrt{\frac{2}{a}}\sin\left(\frac{\pi n}{a}x\right)dx$$

$$= \frac{2\alpha}{a}\sin^2\left(\frac{\pi n}{a}\,\frac{a}{2}\right) = \frac{2\alpha}{a}\sin^2\left(\frac{\pi n}{2}\right)$$

Thus $E_n^1 = \frac{2\alpha}{a}\sin^2\left(\frac{\pi n}{2}\right) = \begin{cases} \frac{2\alpha}{a} & \text{for } n = \text{odd} \\ 0 & \text{for } n = \text{even} \end{cases}$

This is because for n=even ψ_n is <u>odd</u> with respect to the middle point $a/2$. $\Rightarrow$ it does not feel the perturbation which acts exactly at that point!

(b)

$$\psi_n^1 = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{E_n^0 - E_m^0}\,\psi_m^0$$

$$\langle \psi_m^0 | H' | \psi_n^0 \rangle = \frac{2\alpha}{a}\sin\left(\frac{\pi n}{2}\right)\sin\left(\frac{\pi m}{2}\right)$$

$\Rightarrow$ for $n=1$ (ground state) $\langle \psi_m^0 | H' | \psi_1^0 \rangle = \frac{2\alpha}{a}\sin\left(\frac{\pi m}{2}\right)$

$$\psi_{n=1}^1 = \sum_{m \neq 1} \frac{2\alpha}{a}\,\frac{\sin\left(\frac{\pi m}{2}\right)}{E_1^0 - E_m^0}\,\psi_m^0 = \frac{2\alpha}{a}\left(\frac{\sin\left(\frac{\pi}{2}\cdot 3\right)}{E_1^0 - E_3^0}\psi_3^0 + \frac{\sin\left(\frac{\pi}{2}\cdot 5\right)}{E_1^0 - E_5^0}\psi_5^0 + \frac{\sin\left(\frac{\pi}{2}\cdot 7\right)}{E_1^0 - E_7^0}\psi_7^0\right)$$

Now, $E_1^0 - E_m^0 = \frac{\pi^2\hbar^2}{2ma^2}(1 - m^2)$

$$\Rightarrow \psi_1^1 = \frac{2\alpha}{a}\frac{2ma^2}{\pi^2\hbar^2}\left(\frac{-1}{1-3^2}\psi_3^0 + \frac{+1}{1-5^2}\psi_5^0 + \frac{-1}{1-7^2}\psi_7^0\right)$$

$$= \frac{2\alpha}{a}\frac{2ma^2}{\pi^2\hbar^2}\sqrt{\frac{2}{a}}\left\{\frac{1}{8}\sin\left(\frac{3\pi x}{a}\right) - \frac{1}{24}\sin\left(\frac{5\pi x}{a}\right) + \frac{1}{48}\sin\frac{7\pi x}{a}\right\}$$

$$= \frac{m\alpha}{\pi^2\hbar^2}\sqrt{\frac{a}{2}}\left\{\sin\left(\frac{3\pi x}{a}\right) - \frac{1}{3}\sin\left(\frac{5\pi x}{a}\right) + \frac{1}{6}\sin\left(\frac{7\pi x}{a}\right)\right\}.$$

$\boxed{P6.2}$ $V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2 \Rightarrow E_n^0 = \left(n+\frac{1}{2}\right)\hbar\omega \quad (n=0,1,2\ldots)$

let $k \to (1+\varepsilon)k \Rightarrow \omega = \sqrt{\frac{k}{m}} \to \sqrt{(1+\varepsilon)\frac{k}{m}} = \sqrt{1+\varepsilon}\,\omega$

(a) Using $\sqrt{1+\varepsilon} = 1 + \frac{1}{2}\varepsilon - \frac{1}{8}\varepsilon^2$

$$\Rightarrow E_n = \left(n+\frac{1}{2}\right)\hbar\omega\left(1+\frac{1}{2}\varepsilon - \frac{1}{8}\varepsilon^2\right)$$

(b) On the other hand, we can consider

$H' = \frac{1}{2}(1+\varepsilon)kx^2 - \frac{1}{2}kx^2 = \frac{\varepsilon k}{2}x^2$ as a perturbation.

The first-order correction to $\underline{n}$-th level is

$$E_n^1 = \langle \psi_n^0 | H' | \psi_n^0 \rangle = \frac{1}{2}\varepsilon k \langle \psi_n^0 | x^2 | \psi_n^0 \rangle$$

But $x = \sqrt{\frac{\hbar}{2m\omega}}(a_+ + a_-) \Rightarrow \langle n | x^2 | n \rangle = \frac{\hbar}{2m\omega}$.

$$\langle n | (a_+ + a_-)^2 | n \rangle = \frac{\hbar}{2m\omega}\langle n | a_+a_+ + a_-a_- + a_+a_- + a_-a_+ | n \rangle$$

$$= \frac{\hbar}{2m\omega}\langle n | a_+a_- + a_-a_+ | n \rangle$$

because $a_+a_+|n\rangle \sim |n+2\rangle$

$a_-a_-|n\rangle \sim |n-2\rangle$

But $a_+ \psi_n = \sqrt{n+1}\, \psi_{n+1} \Rightarrow a_+ |n\rangle = \sqrt{n+1}\, |n+1\rangle$

$a_- \psi_n = \sqrt{n}\, \psi_{n-1} \Rightarrow a_- |n\rangle = \sqrt{n}\, |n-1\rangle$ eq. 2.66

$$\Rightarrow \langle n| a_+ a_- |n\rangle = \langle n| a_+ \sqrt{n}\, |n-1\rangle =$$

$$= \sqrt{n}\, \langle n| \sqrt{n-1+1}\, |n\rangle = \sqrt{n}\cdot\sqrt{n}\, \langle n|n\rangle = n$$

and $\langle n| a_- a_+ |n\rangle = \langle n| a_- \sqrt{n+1}\, |n+1\rangle =$

$$= \sqrt{n+1}\, \langle n| a_- |n+1\rangle = \sqrt{n+1}\, \langle n| \sqrt{n+1}\, |n+1-1\rangle =$$

$$= \left(\sqrt{n+1}\right)^2 \langle n|n\rangle = n+1$$

so that $\langle n| x^2 |n\rangle = \frac{\hbar}{2m\omega}\left(n+1+n\right) = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)$

$$\Rightarrow E_n^1 = \frac{1}{2}\varepsilon k \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right) = \frac{1}{2}\varepsilon \cdot m\omega^2 \cdot \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right) =$$

$k = m\omega^2$

$$= \frac{1}{2}\varepsilon\, \hbar\omega\left(n+\frac{1}{2}\right) = \frac{1}{2}\varepsilon \cdot E_n^0$$

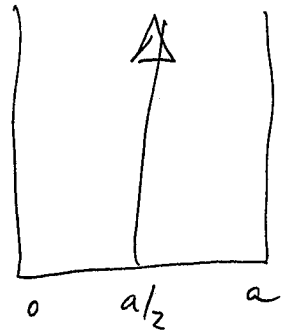
Comparing with linear-ε term in (a):

$$E_n = \underbrace{\left(n+\frac{1}{2}\right)\hbar\omega}_{E_n^0}\left(1+\frac{1}{2}\varepsilon\right) = E_n^0\left(1+\frac{1}{2}\varepsilon\right)$$

we see that perturbation theory gives exactly the same correction:

$$E_n^0 + E_n^1 = E_n^0\left(1+\frac{1}{2}\varepsilon\right) = \hbar\omega\left(n+\frac{1}{2}\right)\left(1+\frac{1}{2}\varepsilon\right).$$

$\boxed{P6.4}$ (a) $H' = \alpha\, \delta(x - a/2)$



$$E_n^0 = \frac{\pi^2 \hbar^2}{2ma^2} n^2$$

Matrix elements: $\langle \psi_m^0 | H' | \psi_n^0 \rangle =$

$$= \int_0^a dx \sqrt{\frac{2}{a}} \sin\left(\frac{\pi m}{a} x\right) \alpha\, \delta(x - a/2) \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n}{a} x\right) =$$

$$= \alpha \cdot \frac{2}{a} \sin\left(\frac{\pi m}{a} \frac{a}{2}\right) \sin\left(\frac{\pi n}{a} \frac{a}{2}\right) = \frac{2\alpha}{a} \sin\left(\frac{\pi m}{2}\right) \sin\left(\frac{\pi n}{2}\right)$$

$\Rightarrow$ finite only for $m = \text{odd}$, $n = \text{odd}$

(n=even states do not "feel" $\delta(x - a/2)$ because wave function is zero there!)

Note that $|\langle \psi_m^0 | H' | \psi_n^0 \rangle|^2 = \left(\frac{2\alpha}{a}\right)^2 \delta_{m,\text{odd}}\, \delta_{n,\text{odd}}$

$$\Rightarrow E_n^2 = \left(\frac{2\alpha}{a}\right)^2 \sum_{m \neq n}{}' \frac{\delta_{m,\text{odd}}\, \delta_{n,\text{odd}}}{E_n^0 - E_m^0} =$$

$$= \frac{4\alpha^2}{a^2} \frac{2ma^2}{\pi^2 \hbar^2} \sum_{m \neq n}{}' \frac{\delta_{m,\text{odd}}\, \delta_{n,\text{odd}}}{n^2 - m^2} =$$

$$= \frac{8m\alpha^2}{\pi^2 \hbar^2} \sum_{m \neq n}{}' \left(\frac{1}{m+n} - \frac{1}{m-n}\right) \frac{1}{2n}$$

Consider $\underline{n=1}$: $\frac{1}{2} \sum_{m=\text{odd}}{}' \left(\frac{1}{m+1} - \frac{1}{m-1}\right) = \frac{1}{2} \left\{ \left(\overset{m=3}{\frac{1}{4}} + \overset{5}{\frac{1}{6}} + \overset{7}{\frac{1}{8}} + \ldots \right) \right.$

$$\left. - \left(\overset{m=3}{\frac{1}{2}} + \overset{5}{\frac{1}{4}} + \overset{7}{\frac{1}{6}} + \ldots \right) \right\} = \frac{1}{2} \left(-\frac{1}{2}\right) = -\frac{1}{4}$$

Consider $\underline{n=3}$ (there is $\underline{\text{no}}$ correction for $n=2=$ even!)

$$\frac{1}{6}\sum_{\substack{m=\text{odd}\\ m\neq 3}}{}' \left(\frac{1}{m+3}-\frac{1}{m-3}\right) = \frac{1}{6}\left\{\left(\overset{m=1}{\frac{1}{4}}+\overset{5}{\frac{1}{8}}+\overset{7}{\frac{1}{10}}+\dots\right)\right.$$

$$\left.-\left(-\overset{m=1}{\frac{1}{2}}+\overset{5}{\frac{1}{2}}+\overset{7}{\frac{1}{4}}+\overset{9}{\frac{1}{6}}+\overset{11}{\frac{1}{8}}+\dots\right)\right\} = \frac{1}{6}\left(-\frac{1}{6}\right) = -\frac{1}{36}$$

Thus there is a system:

$$\frac{1}{2n}\sum_{m\neq n}{}' \left(\frac{1}{m+n}-\frac{1}{m-n}\right) = \frac{1}{2n}\left(-\frac{1}{2n}\right)$$

This can be seen as follows:

unrestricted sum

$$\frac{1}{2n}\sum_{m\neq n}{}' \frac{1}{m+n} = \frac{1}{2n}\left(\sum_m \frac{1}{m+n} - \frac{1}{n+n}\right)$$

Thus $\frac{1}{2n}\left(\sum_m{}' \frac{1}{m+n} - \sum_m{}' \frac{1}{m-n}\right) =$

$$= \frac{1}{2n}\left(\sum_m \frac{1}{m+n} - \sum_{m\neq n}{}' \frac{1}{m-n} - \frac{1}{2n}\right) = \frac{1}{2n}\left(-\frac{1}{2n}\right)$$

these two $\underline{\text{infinite}}$ sums cancel each other

Hence

$$\begin{cases} E_n^2 = \dfrac{8m\alpha^2}{\pi^2\hbar^2}\cdot\dfrac{-1}{(2n)^2} = -\;2m\left(\dfrac{\alpha}{\pi\hbar n}\right)^2 & \text{for } \underline{n=\text{odd}} \\ E_n^2 = 0 \quad \text{for} \quad n=\text{even}. \end{cases}$$

(b) $$E_n^2 = \sum_{m\neq n}{}' \frac{|\langle m|H'|n\rangle|^2}{E_n^0 - E_m^0} \qquad E_m^0 = \hbar\omega\left(m+\tfrac{1}{2}\right)$$

$$H' = \frac{\varepsilon k}{2} x^2$$

2nd Correction to $n=0$ state:

$$E_0^2 = \sum_{m\neq 0}{}' \left(\frac{\varepsilon k}{2}\right)^2 \frac{|\langle m|x^2|0\rangle|^2}{E_0^0 - E_m^0} = -\left(\frac{\varepsilon k}{2}\right)^2 \frac{1}{\hbar\omega} \cdot$$

$$\cdot \sum_{m\neq 0}{}' \frac{|\langle m|x^2|0\rangle|^2}{m}$$

But $x = \sqrt{\frac{\hbar}{2m\omega}}\,(a_+ + a_-) \Rightarrow x^2 = \frac{\hbar}{2m\omega}(a_+ + a_-)^2$

$$\langle m|(a_+ + a_-)^2|0\rangle = \langle m|\, a_+^2 + a_-^2 + a_+a_- + a_-a_+|0\rangle =$$

$$= \langle m|\, a_+a_+ + a_-a_+|0\rangle = \langle m|\, a_+|1\rangle + \langle m|a_-|1\rangle$$

$$= \langle m|\sqrt{2}|2\rangle + \langle m|0\rangle = \sqrt{2}\,\delta_{m,2} + \delta_{m,0}$$

$$\Rightarrow E_0^2 = -\frac{1}{\hbar\omega}\left(\frac{\varepsilon k}{2}\right)^2\left(\frac{\hbar}{2m\omega}\right)^2 \underbrace{\sum_{m\neq 0}{}' \frac{(\sqrt{2}\,\delta_{m,2})^2}{m}}_{\frac{2}{2}=1} =$$

$$= -\frac{\varepsilon^2 k^2 \hbar}{16 m^2\omega^3} \quad \text{but } k = m\omega^2.$$

$$\Rightarrow E_0^2 = -\frac{\varepsilon^2}{16}\frac{m^2\omega^4}{m^2\omega^3}\hbar = \frac{1}{2}\hbar\omega\left(-\frac{\varepsilon^2}{8}\right)$$

Thus, collecting zeros, first, second corrections:

$$E_0 = \frac{1}{2}\hbar\omega + \frac{1}{2}\hbar\omega\left(\frac{\varepsilon}{2}\right) + \frac{1}{2}\hbar\omega\left(-\frac{\varepsilon^2}{8}\right) =$$

$$= \left(1 + \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8}\right)\frac{1}{2}\hbar\omega.$$

P6.5 $H^0 = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$, $H' = -q\mathcal{E}x$

(a) $E_n^1 = \langle n|H'|n\rangle = -q\mathcal{E}\langle n|\sqrt{\frac{\hbar}{2m\omega}}(a_+ + a_-)|n\rangle =$

$= -q\mathcal{E}\sqrt{\frac{\hbar}{2m\omega}}\left(\langle n|\sqrt{n+1}|n+1\rangle + \langle n|\sqrt{n}|n-1\rangle\right) = 0.$

$\Rightarrow$ no first order correction.

$$E_n^2 = \sum_{m\neq n}{}' (-q\mathcal{E})^2 \frac{|\langle m|x|n\rangle|^2}{E_n^0 - E_m^0} =$$

$$= \frac{q^2\mathcal{E}^2}{\hbar\omega} \sum_{m\neq n}{}' \frac{|\langle m|x|n\rangle|^2}{n-m}$$

$$\langle m|\sqrt{\frac{\hbar}{2m\omega}}(a_+ + a_-)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\langle m|\sqrt{n+1}|n+1\rangle + \right.$$

$$\left. + \langle m|\sqrt{n}|n-1\rangle\right) = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{m,n+1} + \sqrt{n}\,\delta_{m,n-1}\right).$$

$$\Rightarrow E_n^2 = \frac{q^2\mathcal{E}^2}{\hbar\omega}\frac{\hbar}{2m\omega}\sum_{m\neq n}{}' \frac{(n+1)\delta_{m,n+1} + n\,\delta_{m,n-1}}{n-m} =$$

$$= \frac{q^2\mathcal{E}^2}{2m\omega^2}\left(\frac{n+1}{n-(n+1)} + \frac{n}{n-(n-1)}\right) = -\frac{q^2\mathcal{E}^2}{2m\omega^2}$$

does not depend on $\underline{n}$: the same correction for all levels!

(b) $H = H^0 + H' = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - q\mathcal{E}x = \frac{p^2}{2m} + \frac{1}{2}m\omega^2\left(x^2 - \frac{2q\mathcal{E}}{m\omega^2}x\right)$

$$= \frac{p^2}{2m} + \frac{1}{2}m\omega^2\left[\left(x - \frac{q\mathcal{E}}{m\omega^2}\right)^2 - \left(\frac{q\mathcal{E}}{m\omega^2}\right)^2\right] =$$

$$= \frac{p^2}{2m} + \frac{1}{2}m\omega^2(x-x_0)^2 - \frac{q^2\mathcal{E}^2}{2m\omega^2}, \quad x_0 = \frac{q\mathcal{E}}{m\omega^2}$$

Change variables: $x - x_0 = y$; NOTE that $p = \frac{\hbar}{i}\frac{d}{dx} = \frac{\hbar}{i}\frac{d}{dy}$

$$\Rightarrow H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 y^2 - \frac{q^2\mathcal{E}^2}{2m\omega^2}$$

$$\Rightarrow H\psi_n = E_n\psi_n \Rightarrow \left(\frac{p^2}{2m} + \frac{1}{2}m\omega^2 y^2\right)\psi_n = \left(E_n + \frac{q^2\mathcal{E}^2}{2m\omega^2}\right)\psi_n \Rightarrow E_n = \hbar\omega\left(n+\frac{1}{2}\right) - \frac{q^2\mathcal{E}^2}{2m\omega^2}$$

agrees with exact re-formulation